

Dementia Detection from Spontaneous Speech

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Abstract

Alzheimer’s Disease (AD) can impair lexical retrieval, working memory, and discourse organization, all of which can appear in spontaneous speech. We study Probable AD detection on the DementiaBank Pitt Corpus Cookie Theft picture-description task using acoustic, lexical, and discourse-level features. After evaluating HuBERT, WavLM, HuBERT layer variants, TF-IDF, frozen DistilRoBERTa embeddings, and demographic baselines, we propose a three-branch late-fusion system combining TF-IDF word patterns, HuBERT-base acoustic representations, and LLM-scored scene-description quality using development-selected fusion weights. The LLM is used as a constrained scene/discourse annotator rather than as a dementia diagnostician. The final fused system achieves held-out test session AUROC 0.9257, F1 0.8537, accuracy 0.8788, precision 0.8974, and recall 0.8140. Compared with the saved TF-IDF baseline, F1 and accuracy improve significantly under paired testing, while AUROC improves directionally but does not reach statistical significance. On the staff blind evaluation, the submitted fixed-threshold system retained high AUROC 0.9627 and recall 0.9545, but lower accuracy 0.7600 due mainly to false positives, indicating strong ranking performance with recall-heavy threshold calibration. Post-hoc validation shows that final AD probability is stronger than age-only prediction and correlates strongly with MMSE, Spearman $\rho = -0.7376$, despite MMSE not being used as an input.

Index Terms: dementia detection, Alzheimer’s disease, spontaneous speech, HuBERT, TF-IDF, large language models

1. Introduction

Alzheimer’s Disease is the most common form of dementia, yet diagnosis often occurs after meaningful cognitive decline has already developed. Early detection is clinically valuable, but standard screening typically requires in-person assessment by trained clinicians. Automatic analysis of spontaneous speech offers a low-cost and scalable alternative because speech reflects lexical retrieval, attention, memory, and discourse organization.

The Cookie Theft picture-description task, drawn from the Boston Diagnostic Aphasia Examination, is widely used in dementia speech research [1, 2]. Participants describe the same standardized image, so differences in responses are more likely to reflect speaker-specific cognitive and linguistic patterns rather than differences in topic. AD can affect this task in several ways: speakers may omit important scene elements, use vague references, produce fragmented descriptions, or fail to connect events into a coherent description.

In this project, we study AD detection from the DementiaBank Pitt Corpus using the provided train, development,

and test splits. The data contain both audio and cleaned transcripts, enabling acoustic, linguistic, and multimodal modeling. We first reproduce and extend the required HuBERT acoustic and TF-IDF linguistic baselines, including HuBERT layer analysis, WavLM acoustic features, DistilRoBERTa text embeddings, demographic auditing, multimodal fusion, and recall-constrained threshold selection. We then introduce an LLM scene-graph branch that scores observable properties of the Cookie Theft description. Our central hypothesis is that conventional word/audio models capture useful low-level signals, but task-aware scene clarity captures an additional discourse-level signal relevant to dementia detection.

2. Background

Automatic dementia detection from speech has traditionally used linguistic markers such as lexical diversity, part-of-speech patterns, vocabulary specificity, and syntactic complexity [3, 4]. These features reflect clinically meaningful changes in lexical retrieval and discourse production, and help explain why simple TF-IDF models can perform strongly on the Cookie Theft task [5, 6].

Acoustic modeling has increasingly used pretrained self-supervised speech representations such as HuBERT and WavLM [7, 8]. These models provide general-purpose speech embeddings learned from large unlabeled corpora. However, acoustic representations alone may not fully capture the semantic and discourse structure of a picture description. For this reason, multimodal fusion of transcript and audio features is a natural approach.

Recent work has shown that GPT-derived linguistic features can improve Alzheimer’s disease detection from spontaneous speech, motivating the use of LLMs as structured annotators rather than unconstrained diagnosticians [12]. In this setting, the LLM is not asked to decide whether a patient has dementia. Instead, it scores observable properties of the response. Our scene-graph branch follows this design principle by measuring how completely and coherently the participant described the Cookie Theft scene.

3. Project Description

3.1. Task Formulation and Runtime

The DementiaBank Pitt Corpus contains Cookie Theft audio recordings and transcripts from Control and Probable AD participants. The provided data are split into train, development, and test sets. Utterance-level predictions are aggregated to session-level probabilities, and final evaluation is performed at the session level. Model selection is performed using the development set, while the test set is held out for final evaluation. Un-

less otherwise stated, results are reported as session-level AUROC, F1, accuracy, precision, recall, and false-positive/false-negative counts.

Runtime evidence in the repository is cache-based rather than exact wall-clock logging. HuBERT, WavLM, and DistilRoBERTa features were extracted once and reused from cached feature files, while LLM scene-graph scoring was batched, cached, and resumable. The final fusion grid search, threshold selection, and statistical tests operated on saved session-level prediction and score files, so the expensive stages were feature generation and LLM annotation rather than downstream logistic regression or thresholding.

3.2. Text Branches

The main text branch represents each transcript using word 1–2 gram TF-IDF features and trains an L2-regularized logistic regression classifier. This branch is the strongest simple baseline. TF-IDF captures discriminative words and short phrases, including references to Cookie Theft objects and actions. It also indirectly captures vague language patterns through words such as “it,” “there,” and “that.” Despite its simplicity, the saved TF-IDF baseline achieves test AUROC 0.9091, F1 0.7778, and accuracy 0.7980.

We also evaluated frozen DistilRoBERTa transcript embeddings as an alternative neural text representation, motivated by pretrained language-model baselines such as BERT and RoBERTa [9, 10]. This model achieved test AUROC 0.8904, F1 0.7857, and accuracy 0.8182. Although DistilRoBERTa improved hard-label accuracy over the saved TF-IDF baseline, its ranking performance remained below TF-IDF, and it did not improve the final fusion system. We therefore retained TF-IDF as the main linguistic branch because it was strong, simple, and stable under development-set selection.

3.3. Acoustic Branches and HuBERT Layer Sweep

The acoustic branch uses frozen HuBERT-base representations extracted from the participant audio [7]. Segment-level embeddings are mean-pooled and passed to a logistic regression classifier. The saved HuBERT baseline is weaker than the text baseline, obtaining test AUROC 0.7251, F1 0.5854, and accuracy 0.6566.

We completed a HuBERT layer exploration across layers 0–12. For each layer, frozen embeddings were extracted and evaluated using logistic regression, with selection based on development AUROC. Layer 12 was selected by development AUROC and achieved test AUROC 0.7770, F1 0.6512, and accuracy 0.6970. This improved over the saved HuBERT rerun but remained weaker than the TF-IDF text baseline. We also tested WavLM acoustic features, which achieved test AUROC 0.7350, F1 0.6364, and accuracy 0.6768. These acoustic experiments show that pretrained speech embeddings contain useful signal, but they were not sufficient on their own for strong AD detection in this dataset.

3.4. LLM Scene-Graph Branch

The scene-graph branch addresses a limitation of bag-of-words text modeling: two transcripts can contain similar keywords while differing substantially in how clearly they describe the image. We use GPT-4o-mini as a structured annotator. The LLM is not prompted to diagnose dementia. Instead, it scores observable properties of the Cookie Theft description.

The four core dimensions are:

Table 1: *Held-out test session-level results. The final model is selected using development data only.*

Model	AUROC	F1	Acc.
HuBERT-base	0.7251	0.5854	0.6566
WavLM	0.7350	0.6364	0.6768
HuBERT layer-12	0.7770	0.6512	0.6970
DistilRoBERTa	0.8904	0.7857	0.8182
TF-IDF 1–2gram	0.9091	0.7778	0.7980
TF-IDF+HuBERT	0.9140	0.7865	0.8081
TF-IDF+HuBERT+LLM scene	0.9257	0.8537	0.8788

1. **Scene/object coverage:** whether expected elements such as the boy, cookies, stool, sink, and water are mentioned.
2. **Agent-action clarity:** whether it is clear who is performing each action.
3. **Spatial/causal clarity:** whether objects and events are linked into a coherent scene description.
4. **Reference specificity:** whether the speaker uses specific nouns/actions rather than vague references.

The normalized scene-quality scores are used as interpretable features in a logistic regression classifier, producing a session-level scene probability. This design keeps the LLM role constrained: it converts each transcript into task-grounded discourse features rather than directly predicting clinical status.

3.5. Fusion and Threshold Selection

The final system combines three complementary session-level probabilities: a TF-IDF text probability, a HuBERT-base acoustic probability, and an LLM scene-graph probability. Fusion weights were selected by development-set grid search, using development AUROC with development F1 as a tie-breaker. The submitted system first forms a text-acoustic base probability,

$$p_{\text{base}} = 0.9p_{\text{TFIDF}} + 0.1p_{\text{HuBERT}},$$

and then combines this base model with the LLM scene branch,

$$p_{\text{final}} = 0.85p_{\text{base}} + 0.15p_{\text{scene}}.$$

Thus, the effective weights are 0.765 for TF-IDF, 0.085 for HuBERT-base, and 0.15 for the LLM scene branch. The HuBERT component is the saved HuBERT-base final-layer baseline rather than the layer-12 sweep variant.

The submitted operating threshold was 0.519. As a clinically motivated operating-point analysis, we evaluated the recall-constrained criterion of selecting a development threshold that maximizes F1 subject to AD recall ≥ 0.85 . The same 0.519 threshold was selected by this recall-constrained criterion and was then fixed for held-out and blind evaluation.

4. Experiments and Results

4.1. Baseline, Fusion, and Blind Results

Table 1 summarizes the main held-out test session-level results. The additional acoustic experiments improve modestly over the saved HuBERT baseline, but remain weaker than text-only modeling. The TF-IDF branch is a strong baseline, while standard TF-IDF+HuBERT fusion provides a modest improvement in AUROC. The largest hard-label improvement occurs when the LLM scene-graph branch is added.

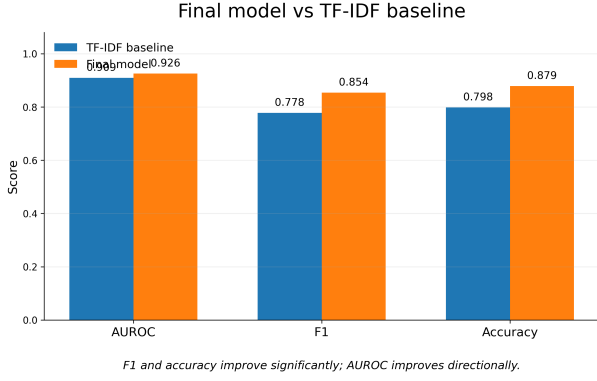


Figure 1: Final model compared with the TF-IDF baseline. F1 and accuracy gains are statistically significant; AUROC improves directionally.

Table 2: Held-out and staff blind session-level results for the final submitted system. Both use threshold 0.519.

Split	<i>n</i>	AUC	F1	Acc.	Prec.	Rec.	FP/FN
Test	99	0.9257	0.8537	0.8788	0.8974	0.8140	4/8
Blind	50	0.9627	0.7778	0.7600	0.6563	0.9545	11/1

Compared with the saved TF-IDF baseline predictions, the final model improves F1 by 0.0759 and accuracy by 0.0808. Paired randomization tests indicate that both improvements are statistically significant ($p = 0.0074$ for F1 and $p = 0.0083$ for accuracy). The AUROC improvement is positive but not statistically significant ($\Delta = 0.0166$, $p = 0.0737$). McNemar’s exact test on hard-label corrections shows that the final model fixes eight TF-IDF errors while introducing no paired hard-label regressions ($p = 0.0078$). Thus, the strongest evidence is for improved operating-point classification rather than a large global ranking shift.

The final submitted system used a fixed threshold of 0.519, selected from the development set. Table 2 compares the held-out test result with the staff blind evaluation. The blind set shows very strong ranking performance, with AUROC 0.9627, but lower hard-label accuracy and F1. The error breakdown shows that this was mainly a threshold-calibration issue: the model produced only one false negative but eleven false positives, giving high recall but reduced precision. No post-hoc threshold recalibration was applied during blind evaluation.

4.2. Statistical Testing

Statistical comparisons with the TF-IDF baseline were performed using paired session-level tests on the 99 held-out test sessions. For AUROC, F1, and accuracy, we used 10,000 paired randomization trials. In each trial, the two models’ outputs were independently swapped within each session, the metric difference was recomputed, and a two-sided p -value was calculated as the fraction of null differences at least as extreme as the observed difference. AUROC was recomputed from paired probabilities, while F1 and accuracy were recomputed from paired hard labels using the final model threshold 0.519 and the saved TF-IDF probabilities thresholded at 0.5. We also computed 10,000 paired bootstrap confidence intervals over sessions and applied McNemar’s exact two-sided binomial test to paired correctness indicators.

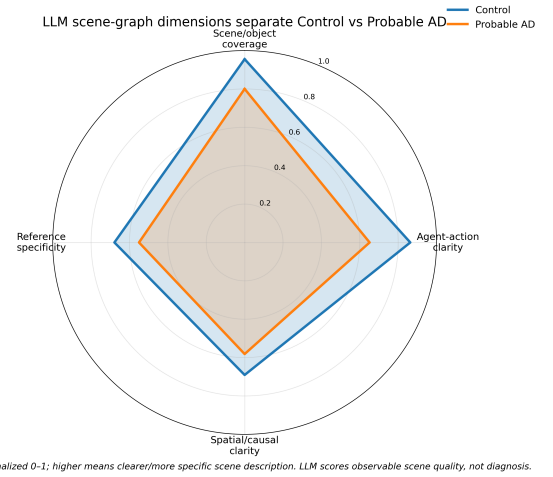


Figure 2: Average normalized LLM scene-graph scores on the test split. Higher scores indicate clearer, more specific scene descriptions. The LLM scores observable description quality rather than diagnosis.

4.3. LLM Scene-Graph Analysis

Figure 2 shows average normalized LLM scene-graph scores for Control and Probable AD sessions on the test set. Control sessions score higher across all four dimensions, suggesting that the scene-graph branch captures meaningful differences in picture-description quality. This supports the interpretation that the LLM branch adds task-specific discourse information beyond generic word counts.

4.4. Ablations and Additional Experiments

Beyond the main branches shown in Table 1, we grouped additional experiments into four families: acoustic/timing features, alternative text representations, LLM-derived semantic/discourse features, and decision-structure variants. Acoustic and timing experiments included ASR-error features, ASR transcript fusion, and pause/disfluency descriptors. Alternative text experiments included frozen DistilRoBERTa embeddings and transcript perturbation aggregation. LLM-derived variants included semantic-specificity features, event triples, and multi-view scene profiles. Decision-structure variants included hard-case adjudication, marker-based correction, cascade rules, and two-pass uncertainty resolution.

These experiments were useful analytically but did not produce a robust improvement over the final forced session-level classifier. The most competitive alternatives were still closely related to the final model. Event-triple and semantic-specificity features captured interpretable aspects of the Cookie Theft description, but largely overlapped with the information captured by the scene-graph branch. In contrast, hard-case adjudication and cascade correction tended to overfit development errors and introduced new test regressions. This pattern suggests that the final scene-graph branch was useful because it added a broad, stable discourse-quality signal, while narrower correction modules were less reliable.

Two exploratory settings did improve performance under modified evaluation assumptions. Subject-prefix aggregation, which combines repeated recordings from the same participant, reached test AUROC 0.9569 and F1 0.8889, but changes the

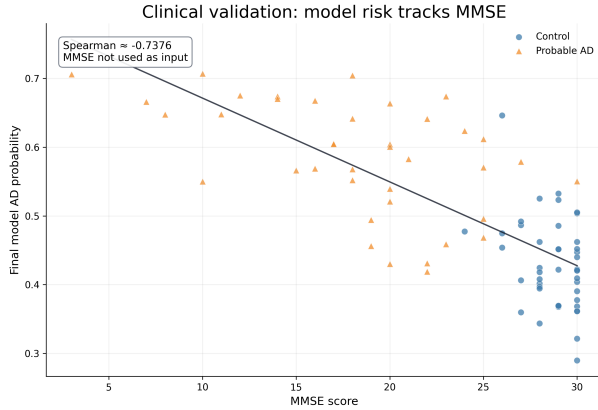


Figure 3: *Clinical validation: final model AD probability is negatively correlated with MMSE score, although MMSE is not used as a model input.*

evaluation unit from session to subject prefix. Abstention/triage experiments also improved classified-case accuracy at reduced coverage, but no longer perform forced classification for every session. We therefore report these as promising extensions rather than as the main result.

4.5. Clinical and Demographic Validation

To assess whether the model captures clinically meaningful signal rather than spurious corpus artifacts, we performed post-hoc validation using participant metadata. These metadata were never used as classifier inputs.

As a demographic audit, we first evaluated whether simple metadata alone could explain the classification results. Age-only prediction was substantially weaker than the final speech/text model, with test AUROC 0.7135, and sex/speech-quantity baselines were weaker still. This suggests that although the corpus contains demographic structure, the final model is not merely reproducing a demographic shortcut.

We then correlated the final model’s predicted AD probability with participant MMSE scores on the held-out test set. MMSE was never used as a classifier input. The final probability correlates strongly and negatively with MMSE across test sessions with available metadata (Spearman $\rho = -0.7376$), and remains negatively correlated within AD-only sessions (Spearman $\rho = -0.4980$). False-negative AD sessions have median MMSE approximately 22, suggesting that the model misses milder cases more often than severe cases.

5. Discussion

The strongest result is that task-aware scene/discourse scoring improves hard-label AD detection beyond conventional acoustic and lexical baselines. TF-IDF is a strong baseline because word and phrase choices differ substantially between Control and Probable AD sessions. However, TF-IDF cannot directly measure whether a participant produced a complete and coherent scene description. The LLM scene-graph branch fills this gap by scoring observable dimensions of picture-description quality.

The broader baseline exploration clarifies the role of each modality. HuBERT, WavLM, and HuBERT layer variants all showed that audio contains signal, but acoustic-only models

were consistently weaker than text-only models on this dataset. DistilRoBERTa showed that neural text embeddings are competitive, but the simpler TF-IDF baseline remained stronger in AUROC and dominated the final development-selected fusion weight. These results suggest that, for this particular Cookie Theft setup, lexical content and discourse completeness dominate over generic acoustic or embedding-based representations.

The held-out and blind results show the difference between ranking performance and fixed-threshold classification. The final model separates the classes well, as reflected by AUROC 0.9257 on the held-out test set and 0.9627 on the staff blind set. However, the blind-set threshold behavior was recall-heavy: the submitted threshold produced only one false negative but eleven false positives. This suggests that the model’s probability ranking transferred well, while its operating threshold was less well calibrated on the blind distribution.

The demographic audit is also important. Age-only prediction was weaker than the final model, supporting that the system is not simply using demographic structure as a shortcut. However, the corpus still contains demographic differences between groups, so demographic metadata should remain part of model auditing even when they are not used as classifier inputs.

A key limitation is the small held-out test set. Although F1 and accuracy improvements are statistically significant, the AUROC improvement is directional but not significant. The blind set is also small, with 50 sessions, so the blind false-positive pattern should be interpreted as an operating-point calibration issue rather than as a complete characterization of deployment performance. Finally, subject-prefix aggregation is promising but exploratory because it changes the evaluation unit from session-level prediction to patient-level prediction.

6. Future Work

Several directions remain open. First, external validation on additional dementia speech datasets is necessary to test whether the scene-graph features generalize beyond the Cookie Theft task. Second, richer acoustic timing features should be extracted from word- or segment-level timestamps to better capture pause behavior and fluency. Third, probability calibration and operating-threshold selection should be studied more carefully, especially because the blind evaluation showed strong AUROC but recall-heavy hard-label behavior at the submitted threshold.

Fourth, patient-level modeling is a promising direction for participants with repeated recordings. Our exploratory subject-prefix aggregation improved stability and reached stronger patient-level performance, but it changes the evaluation unit from session to participant and was therefore not used as the main result. Future work could formalize this setting by learning how to aggregate multiple sessions per patient, or by using carefully controlled transcript/audio augmentations to simulate multiple views of the same participant while avoiding label leakage or artificial clinical evidence.

Finally, the abstention analysis suggests a clinically realistic triage workflow in which high-confidence cases are classified automatically while uncertain cases are referred for clinician review.

7. Conclusion

We presented a three-branch fusion system for automatic AD detection from Cookie Theft picture descriptions. By combining TF-IDF word patterns, HuBERT acoustic representations,

and LLM-scored scene description quality, the system achieves held-out test session AUROC 0.9257, F1 0.8537, and accuracy 0.8788. The key contribution is the LLM scene-graph branch, which captures discourse-level clarity as an interpretable, task-grounded signal. The required baseline explorations showed that HuBERT layer selection, WavLM, and DistilRoBERTa were useful comparisons but did not outperform the final fusion system. Significant F1 and accuracy improvements over the TF-IDF baseline, together with demographic auditing and strong correlation between predicted AD probability and MMSE, support the value of combining conventional speech/text models with structured scene-understanding features. On the staff blind evaluation, the system retained very high AUROC 0.9627, but the fixed threshold produced a recall-heavy operating point with high recall and reduced precision, highlighting calibration as an important direction for future deployment.

8. References

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